CYBERSAATHI

Technical Design and Architecture Document

1. Project Overview

CyberSaathi is a smart digital safety and cyber fraud awareness platform designed to help users identify potentially suspicious links and understand online threats.

The application provides risk alerts, cyber safety tips, and awareness about phishing, fake KYC messages, and online scams.

2. System Architecture

CyberSaathi follows a simple client-server architecture consisting of three main layers:

1. Presentation Layer (Frontend)

- Developed using HTML, CSS, and JavaScript.
- Provides a user-friendly interface.
- Allows users to enter suspicious URLs.
- Displays high-risk or low-risk alerts.
- Provides Check Safety and Clear buttons.

2. Application Layer (Backend)

- Developed using Java.

- Receives URL analysis requests from the frontend.

- Applies URL validation and risk assessment rules.

- Returns risk results and safety recommendations.

3. Data and Awareness Layer

- Contains cyber safety tips and awareness information.

- Uses sample URLs and test cases for demonstration.

- Can be extended with threat intelligence databases.

3. System Architecture Diagram

<CodeBlock language="mermaid" model_previewable>

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flowchart TD
    A[User] --> B[CyberSaathi Web Interface]
    B --> C[Enter Suspicious URL]
    C --> D[Java Backend]
    D --> E[URL Validation]
    E --> F[Risk Assessment Engine]
    F --> G{Risk Level}
    G -->|High Risk| H[High-Risk Alert]
    G -->|Low Risk| I[Low-Risk Alert]
    H --> J[Safety Recommendations]
    I --> J
```

J --> K[User Awareness]

</CodeBlock>4. Technologies Used

Component| Technology

Frontend| HTML

Styling| CSS

Interactivity| JavaScript

Backend| Java

Development| Visual Studio Code

Version Control| Git and GitHub

Testing| Manual functional testing

5. Working Process

1. The user opens the CyberSaathi application.

2. The user enters a URL in the Link Checker.

3. The frontend validates the input.

4. The URL is sent to the Java backend for analysis, if backend integration is enabled.

5. The risk assessment logic checks the URL against configured detection rules.

6. The application displays a high-risk or low-risk alert.

7. The user receives appropriate cyber safety guidance.

8. The Clear button resets the input and results.

6. Risk Assessment Logic

The application uses configured detection rules to identify potentially suspicious URLs.

High-Risk Indicators

- Suspicious or misleading domain names.
- Unknown or unusual URL patterns.
- Phishing-related keywords.
- URLs matching configured suspicious patterns.

Low-Risk Indicators

- URLs matching configured trusted or safe demonstration criteria.
- No suspicious indicators detected by the implemented rules.

Note: A low-risk result does not guarantee that a website is completely safe.

7. Functional Modules

Module| Purpose

Link Checker| Accepts and analyzes URLs

Risk Assessment| Identifies potential threats

Alert System| Displays risk notifications

Awareness Module| Provides cyber safety tips

Clear Function| Resets user input and results

8. Security Considerations

- Do not request or store users' passwords, OTPs, or banking credentials.
- Validate user input before processing.
- Avoid automatically opening suspicious URLs.
- Use secure communication (HTTPS) when deployed.
- Do not store personal information unnecessarily.
- Clearly explain that risk alerts are based on implemented detection rules.

9. Testing and Validation

Test Case	Expected Result
Suspicious sample URL	High-risk alert
Safe demonstration URL	Low-risk alert
Empty URL input	Validation message
Invalid URL format	Invalid URL message
Clear button	Input and result cleared
Check Safety button	Risk assessment executed

10. Future Enhancements

- AI-based phishing detection.
- Integration with trusted threat intelligence APIs.
- Real-time malicious URL database.
- Support for multiple Indian languages.

- Mobile application development.

- Cybercrime reporting assistance.

11. Conclusion

CyberSaathi provides an accessible approach to cyber safety awareness by combining URL risk assessment, alerts, and educational guidance. The project aims to empower citizens with the knowledge and tools needed to make safer digital decisions.

Project: CyberSaathi

Theme: Digital Safety & Cyber Fraud Awareness

Team Size: 3 Members